

Deepika SINGH

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PROFESSIONAL EXPERIENCE

- Research Engineer** Jul 2024 – Present
Centre de Mathématiques Appliquées (CMAP), École Polytechnique Palaiseau, France
– Design advanced Machine Learning (ML) methods for fault detection, diagnostics, and predictive maintenance (PHM).
– Optimize industrial processes by integrating multi sensor datasets, with a focus on battery health management.
- Research Engineer / Postdoctoral Researcher** Jul 2022 – Jun 2024
Commissariat à l'énergie atomique et aux énergies alternatives (CEA) LIST Paris-Saclay, France
– Developed decentralized and robust federated learning algorithms for secure distributed AI.
– Engineered countermeasures against backdoor and poisoning attacks, strengthening AI trustworthiness in industrial applications.
- Scientific Researcher** Nov 2019 – Jan 2021
Medizinische Universität Graz (MedUni) Graz, Austria
– Contributed to the EU-funded *FeatureCloud* project, advancing privacy-preserving healthcare AI.
– Interpretable and Explainable ML models to enhance transparency in medical domain.
- Junior Scientist (Marie Curie Ph.D Fellow)** Sep 2016 – Oct 2019
Austrian Institute of Technology (AIT) GmbH Wiener Neustadt, Austria
– Marie Skłodowska-Curie fellow in H2020 *ACROSSING*, Machine learning and Deep learning for smart homes and Ambient Assisted Living (AAL).
– Developed models for user behaviour, activity recognition, privacy-preserving data management and dialogue systems in AAL environments
– Visiting Researcher: *De Montfort University, UK, Apr 2018- May 2018*,
Zuyderland Nursing Homes, Netherlands, Apr 2017 - May 2017
- Lecturer & Research Fellow** Aug 2014 – Jun 2015
Malaviya National Institute of Technology (MNIT) Jaipur, India
– Taught courses in Image Processing, Data Mining, and Algorithm Design for bachelor's students.
– Conducted research activities in collaboration with Master's and Ph.D students.
- Lecturer** Jun 2011 – Jul 2012
PCE Engineering college Jaipur, India
– Taught course of C Programming Language for bachelor's students.
– Conducted technical activities among students for software development.

EDUCATION

- PhD in Computer Science** Nov 2017 – Jul 2021
Graz University of Technology (TU Graz) Graz, Austria
– Thesis: *Machine Learning-based Smart Home Systems in Ambient Assisted Living*.
– Advisor: *Prof. Andreas Holzinger, TU Graz*
- Masters (M.Tech.) in Computer Science** Jul 2012 – May 2014
Banasthali University Rajasthan, India
– Thesis: *Early Detection of Malignant Melanoma* using machine learning techniques.
– Advisor: *Prof. Mustaq Ahmed, MNIT*
- Bachelors (B.Tech.) in Computer Engineering** Jul 2007 – May 2011
Rajasthan Technical University Rajasthan, India
– Thesis: *Website Development and Payroll Management System* using ASP.NET.

TECHNICAL SKILLS

Machine Learning/AI: Deep Learning, Explainable AI, Computer Vision, Image Processing, Federated Learning, Natural Language Processing, Predictive Maintenance
Frameworks & Tools: TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, SHAP, Pandas, Seaborn
Programming: Python, R, MATLAB, C, C++
Other: Project management, Research supervision, Academic writing

ACHIEVEMENTS

- Best student paper award at TPS-ISA 2023
Fuba: Federated uncovering of backdoor attacks for heterogeneous data
- Best student paper award at ICOST 2018
Users' perceptions and attitudes towards smart home technologies
- Awarded Marie Skłodowska-Curie Fellowship by the European Commission
Horizon 2020, MSCA-ITN Grant 2016-2019
- Best presenter at ISCA 2013
Three dimensional computer processors
- Outstanding engineering student (semester topper 2009-2010)
Department of Computer Science and Engineering, ACERC, RTU

TALKS AND POSTERS

- Battery Intelligence, AABC Europe 2025, Mainz, Germany
Degradation-Directed Analysis of Lead-acid Battery from Operational Data: Current State and Challenges
- Seminar: CEA LIST Days 2024, Paris-Saclay
Federated uncovering of backdoor attacks for heterogeneous data
- Webinar 2018: Community Care Smart AT Queensland, Australia
Users' perceptions and attitudes towards smart home technologies
- AIT Ph.D winter school 2017, Austria
Activity recognition in Smart homes
- MCAA General Assembly 2017, Salamanca, Spain
Smart Homes for Ambient Assisted Living Technologies
- ARES 2017, Italy
Human activity recognition using recurrent neural networks
- ARES 2017, Italy
Ambient assisted living technologies from the perspectives of older people and professionals
- ICSPCT 2014, India
Detection techniques for melanoma diagnosis: A performance evaluation

SELECTED PUBLICATIONS

Google Scholar [Publications: 15 Citations: 830 h-index: 12]

- D. Singh, E. Merdivan, J. Kropf, M. Geist, A. Holzinger, and S. Hanke, “**Class imbalance in multi-resident activity recognition: an evaluative study on explainability of deep learning approaches,**” *Universal Access in the Information Society*, vol. 24, pp. 1173–1191, 2025. DOI: 10.1007/s10209-024-01123-0.
- F. Espinoza, D. Singh, A. Mayoue, and C. Gouy-Pailler, “**Défense contre les attaques par porte dérobée en apprentissage fédéré par estimation du motif d’attaque et élagage,**” in *Gretsi 2023 Conference*, Aug. 2023.
- D. Singh, I. Psychoula, J. Kropf, S. Hanke, and A. Holzinger, “**Users’ perceptions and attitudes towards smart home technologies,**” in *Lecture Notes in Computer Science*, Springer International Publishing, 2018, pp. 203–214. DOI: 10.1007/978-3-319-94523-1_18.
- D. Singh, J. Kropf, S. Hanke, and A. Holzinger, “**Ambient assisted living technologies from the perspectives of older people and professionals,**” in *Lecture Notes in Computer Science*, Springer International Publishing, 2017, pp. 255–266. DOI: 10.1007/978-3-319-66808-6_17.
- D. Singh, E. Merdivan, S. Hanke, J. Kropf, M. Geist, and A. Holzinger, “**Convolutional and recurrent neural networks for activity recognition in smart environment,**” in *Towards Integrative Machine Learning and Knowledge Extraction*, Springer International Publishing, 2017, pp. 194–205. DOI: 10.1007/978-3-319-69775-8_12.
- D. Singh, E. Merdivan, I. Psychoula, J. Kropf, S. Hanke, M. Geist, and A. Holzinger, “**Human activity recognition using recurrent neural networks,**” in *Lecture Notes in Computer Science*, Springer International Publishing, 2017, pp. 267–274. DOI: 10.1007/978-3-319-66808-6_18.

LANGUAGES

English (Fluent), French (A2), German (A1)